Given $P(-2,3)$ divides AB internally in the ratio 1:6.

By section formula, $P\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right) = P(-2,3)$

$$P\left(\frac{1(x_2)+6(-3)}{1+6}, \frac{1(y_2)+6(5)}{1+6}\right) = P(-2,3)$$

Equating the coordinates

$$\frac{x_2-18}{7} = -2 \qquad \frac{y_2+30}{7} = 3$$
$$x_2 - 18 = -14 \qquad y_2 + 30 = 21$$
$$x_2 = 4 \qquad y_2 = -9$$

Therefore, the coordinate of B is $(4,-9)$

Exercise 5.4

1. Find the coordinates of the point which divides the line segment joining the points $A(4,-3)$ and $B(9,7)$ in the ratio 3:2.
2. In what ratio does the point $P(2,-5)$ divide the line segment joining $A(-3,5)$ and $B(4,-9)$.
3. Find the coordinates of a point P on the line segment joining $A(1,2)$ and $B(6,7)$ in such a way that $AP = \frac{2}{5}AB$.
4. Find the coordinates of the points of trisection of the line segment joining the points $A(-5,6)$ and $B(4,-3)$.
5. The line segment joining $A(6,3)$ and $B(-1,-4)$ is doubled in length by adding half of AB to each end. Find the coordinates of the new end points.
6. Using section formula, show that the points $A(7,-5)$, $B(9,-3)$ and $C(13,1)$ are collinear.
7. A line segment AB is increased along its length by 25% by producing it to C on the side of B. If A and B have the coordinates $(-2,-3)$ and $(2,1)$ respectively, then find the coordinates of C.

5.7 The Coordinates of the Centroid

Consider a $\varnothing ABC$ whose vertices are $A(x_1,y_1)$, $B(x_2,y_2)$ and $C(x_3,y_3)$.

Let AD, BE and CF be the medians of the $\varnothing ABC$.

The mid-point of BC is $D\left(\frac{x_2+x_3}{2}, \frac{y_2+y_3}{2}\right)$

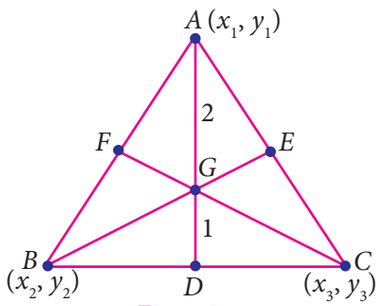


Fig. 5.37

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